

# When is one field enough?

**A measurement audit of an organ-on-a-chip quality-control benchmark, a corrected evaluation protocol, and a chip-level QC tool with an explicit "inconclusive" outcome.**

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|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Team              | <b>solo ranker</b> (individual)                                                                                                                                                                                                                                                                  |
| Member            | Taewoong Kim (independent researcher, South Korea)                                                                                                                                                                                                                                               |
| Kaggle            | <code>aleaiest</code>                                                                                                                                                                                                                                                                            |
| Code, video, demo | <a href="https://github.com/Foreist/when-is-one-field-enough">https://github.com/Foreist/when-is-one-field-enough</a> · video <a href="https://taewoong23-ooc-chip-qc-demo.static.hf.space/video.html">https://taewoong23-ooc-chip-qc-demo.static.hf.space/video.html</a> · browser demo in §6.8 |

## Summary

Organ-on-a-chip (OoC) cultures are inspected daily under a microscope, and the quality-control (QC) decision — keep the chip, re-image, or discard it — is still made by eye. A public benchmark exists for automating this: the **OOC Image Dataset** (3,072 brightfield fields, 59 acquisition sessions, six cell lines, labels from four expert cell biologists by majority vote). Before building on it we asked a simple question: *is this benchmark measuring what it claims to measure?*

It is not, in three specific and measurable ways.

- 1. The published split leaks sessions.** Training and test images come from the *same 57 of 59 sessions*. Holding the test images and the training-set size fixed, and changing only whether the test sessions also contribute training images, moves accuracy by **+7.9 pp (95% CI 4.2-11.5)** and **AUC by +8.9 pp (95% CI 6.7-11.0)** across eight seeds. The shipped split reports 78.9%; the same model on unseen sessions reports 56.0%.
- 2. The 3,072 labels are not 3,072 independent observations.** Labels are strongly autocorrelated along the acquisition order (session ICC 0.321, mean run length 6.08 fields versus 2.03 under independence). The design effect is 17, so the benchmark carries the information of roughly **181 independent labels** — and a single field agrees with the session's majority label only **78.8%** of the time (91.0% even with ten fields).
- 3. The failure mode is spatial, not per-image — and our own policy comparison did not survive its own criticism.** Failures occupy contiguous stretches of a chip (runs test significant in 23/45 mixed sessions, surviving a cell-type control). A policy comparison simulated on the *labels* suggests random fields are best for the chip call; **with the model in the loop that ranking disappears** — over 13–15 chips no paired difference survives a correction for the six comparisons made (§4.3). What does survive is a concrete, reproducible failure: reading the **first  $k$  fields** called a 100%-bad chip *pass* with 0.94 confidence, because the start of a session can sit inside a good region. The shipped tool samples fields spread across the chip for that reason, not on the strength of a policy ranking.
- 4. The leakage is not unique to this benchmark.** Auditing a second public organoid imaging benchmark (OCT organoid tracking, zenodo.15783866) shows the same defect: **40.0% of its test files belong to a (well, day) acquisition group that also appears in training** (§4.4).

We then built the tool the corrected protocol implies, and measured what it buys: on 25 unseen chips it reaches the same chip-level accuracy as reading **every** field (0.800) while using **9.5 fields instead of 27.4** — a 2.9× reduction in microscope time — and defers 8% of chips to a human rather than guessing. Fields are sampled **spread across the chip** (reading the first  $k$  can sit inside a good region and stop early — we observed a 100%-bad chip called *pass* with 0.94 confidence), and a sequential stopping rule returns **pass / fail / inconclusive** with a posterior confidence. On 25 unseen chips: **82.6% chip-level accuracy on the chips it calls (23 of 25), 13% of calls confident-but-wrong, 8% inconclusive, 9.5 fields on average**. A plain cap of 12 spread fields per chip does as well on field count (9.0 on average, 0.800); the saving comes from not reading every field, and what the sequential rule adds is a posterior confidence and the option to defer. The 8-field minimum behind these numbers was tuned on the same 25 chips; re-selected by cross-validation on the other 34 sessions it would be 1, which scores 0.680 here (§6.2(c)). On those independent chips the rule still matches reading every field with a fifth to two-fifths of the fields.

We also report two things we could not do, because they define the honest boundary of this work: the **re-image vs discard** distinction is *not* supported by the data (a constructed recovery target is not predictable; CV AUC 0.613 against a permutation null of 0.564), and **no reliable out-of-distribution detector** was found (image statistics and feature-space distance both missed the worst failure). Finally, simulating a stopping rule on ground-truth labels **overstates** it: on the same 25 chips, without a minimum-field guard, labels give 0.875 and the model 0.708.

Everything is reproducible from one public dataset with the scripts in this repository.

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## 1. Introduction

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### 1.1 The problem

Microphysiological systems — organ-on-a-chip, organoids in microfluidic devices — are promoted as human-relevant alternatives to animal testing. Their practical bottleneck is not the biology but the *bookkeeping*: a chip is imaged repeatedly over days, and at each session someone must decide whether the field in front of them is usable. Technical artifacts (air bubbles, defocus, deformed channel walls) and biological problems (cells not attaching, density far from the expected range) both appear as "bad", and both are judged by eye, field by field.

Automating this judgement is attractive, and a public benchmark for it exists. But an automated QC system inherits every property of its benchmark: if the benchmark's split leaks, reported accuracy is inflated; if its labels are clustered, its effective sample size is far smaller than it looks; if its labels are field-dependent, then "accuracy per image" is not the quantity a lab actually needs. None of these properties are usually checked.

**Who this is for.** (i) *Chip and organoid labs* that image cultures daily and decide keep / re-image / discard by eye; (ii) *imaging and data teams* building reusable, trustworthy culture datasets; (iii) *assays downstream of QC* — dose-response, toxicity, drug evaluation — which inherit the quality of the fields that were kept.

### 1.2 What this report does

We take one public OoC QC benchmark — the OOC Image Dataset [1,2] — and treat the benchmark itself as the object of study. We (i) audit its split and label structure, (ii) derive and validate a corrected evaluation protocol, (iii) measure how sampling and aggregation policies behave on the measured structure, and (iv) build a chip-level QC tool and evaluate it on held-out sessions, reporting which of its settings were chosen on the test chips (§6.2(c)).

## 1.3 What we claim — and what we do not

We claim **measurement results** about a specific benchmark and a **protocol/tool** that follows from them. We do *not* claim a new model architecture, biological or clinical validity, or generalisation beyond this dataset. Leakage and label clustering are known phenomena in machine learning [7–11]; our contribution is not their discovery but their **measured magnitude in this benchmark, the corrected protocol, and the field-sampling policy we adopted after one observed failure** — together with two negative results that bound the claims. The policy is not a measured optimum: a ranking of sampling policies did not survive the model in the loop (§4.3).

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## 2. Data

**Source.** OOC Image Dataset, Zenodo record [10.5281/zenodo.10203721](https://zenodo.org/record/10203721) [3]; data descriptor in *Data* 2024 [1] and a companion conference paper [2]. The Zenodo record states **CC-BY-4.0**; the MDPI descriptor lists **CC-BY-SA**. We resolve the ambiguity conservatively: we redistribute no images beyond the 44 attributed demo example fields, license our code MIT and our derived artefacts CC-BY-SA.

**Content.** 3,072 brightfield images ( $2,056 \times 1,542$  px) from an automated microscope on an OoC setup, covering six cell lines (A549, Caco-2, HPMEC, HUVEC, NHBE, HSAEC). Filenames are `YYMMDD_N.png`: YYMMDD identifies an **acquisition session** (59 in total) and N the field number within it. Each image carries a binary label, `good` or `bad`, assigned by **four expert cell biologists**, with disagreement resolved by majority vote or an additional rater [1]. Metadata (cell type, seeding density, flow rate, day after seeding) is available for part of the data. Figure 1 summarises the corpus: sessions range from 6 to 229 fields (median 24), and the share of fields labelled `bad` ranges from 0% to 100% per session.

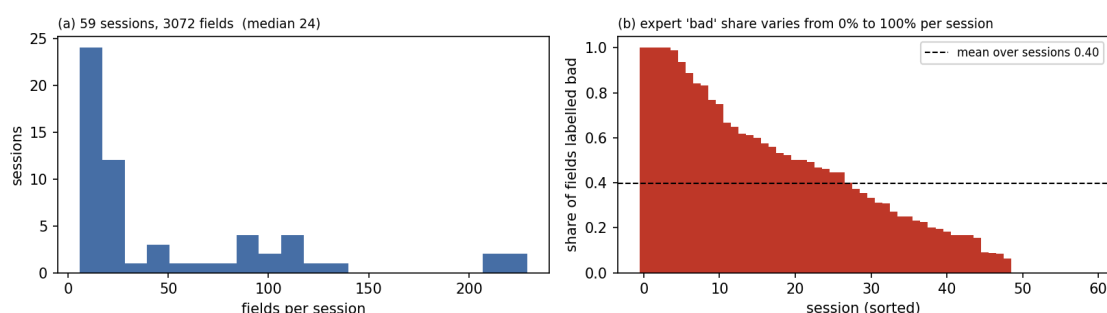


Figure 1. The corpus. (a) fields per session; (b) per-session share of fields labelled `bad`.

**The published split.** The zip ships `train/val/test` folders (2,130 / 286 / 656 images). This is the split used by the reference baseline in [1].

**Privacy, ethics and compliance.** The dataset contains brightfield images of immortalised cell lines grown in microfluidic devices. There are no human subjects, no patient-derived material, no identifiable personal data and no clinical records, so no consent or IRB review applies; the data are distributed publicly under the licences above. The tool automates a quality judgement about an image — it makes no biological, diagnostic or clinical claim, and we do not use it to make decisions about patient material. The second benchmark audited in §4.4 is likewise public (CC-BY-4.0) and contains only instrument images.

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### 3. Methods

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All quantities below are defined once here and used unchanged throughout; the scripts that compute them are listed in §10.

**Sessions and fields.** A *session* is one acquisition date (YYMMDD in the filename) — one microscope run over one or more chips. A *field* is one image. We treat the field index  $N$  as the acquisition order; the proxy is supported by the high correlation between consecutive fields (§4.2). A *run* is a maximal contiguous block of equal labels in field order.

**Runs test (clustering).** For a binary sequence with  $n_1$  bad and  $n_0$  good fields, the number of runs  $R$  has, under independence, mean  $\mu = 2n_1n_0/n + 1$  and variance  $\sigma^2 = 2n_1n_0(2n_1n_0 - n) / (n^2(n - 1))$ ; we report  $z = (R - \mu)/\sigma$  and the two-sided normal p-value. Negative  $z$  means fewer runs than chance, i.e. clustering. Sessions without both labels are skipped. The i.i.d. expected run length is  $1 / (2p(1-p))$  with  $p$  the observed good fraction.

**Intraclass correlation and design effect.** For clusters (sessions) with sizes  $n_i$ , ICC(1) is estimated by one-way ANOVA,  $ICC = (MSB - MSW) / (MSB + (m_0 - 1) \cdot MSW)$  with  $m_0 = (N - \sum n_i^2 / N) / (k - 1)$ ; the design effect is  $1 + (m_0 - 1) \cdot ICC$  and the effective sample size is  $N / \text{design effect}$ . This is the standard cluster-sampling treatment of "images within a chip".

**Controlled leakage A/B.** Test images are fixed by choosing 20 sessions and withholding half of each session's fields as the test set. The other half of those sessions' fields are *available* but used only in the leaky arm. Both arms receive exactly  $N$  training fields ( $N$  = the non-test pool minus the 200 shared validation fields): the disjoint arm draws all  $N$  from non-test sessions; the leaky arm draws  $N - k$  from non-test sessions and  $k$  from the available test-session fields. Model, budget, augmentation and seeds are identical.

**Chip-level decision.** For a set of fields with per-field bad probabilities  $p_j$ , the sequential rule maintains a Beta(1,1) posterior on the bad fraction  $f$  —  $f \mid \text{data} \sim \text{Beta}(1 + b, 1 + g)$  with  $b, g$  the counts of fields with  $p_j > 0.5$  and  $\leq 0.5$  — and stops as soon as  $P(f > 0.5) > 0.9$  or  $P(f < 0.5) > 0.9$ , at most after a budget of 20 fields and never before `min_fields`. If the budget is exhausted with  $0.35 < P(f > 0.5) < 0.65$ , the call is **inconclusive** rather than forced. The confidence 0.9 and budget 20 were fixed in the label-only simulation (`audit/stopping_rule.py`) before any model existed, and the  $\pm 0.15$  band is a round number that was never tuned; none of the three was varied on the test chips. `min_fields` was, and §6.2 reports how it was re-selected without them.

**Spread-field order.** With  $n$  available fields and a budget of  $k$ , the rule inspects indices `round(i · (n-1)/(k-1))` for  $i = 0 \dots k-1$ , i.e. fields spaced evenly across the acquisition order. This replaces "the first  $k$  fields" after the failure described in §6.2.

**Metrics.** *Field accuracy* is the fraction of fields whose predicted side of 0.5 matches the expert label. *Chip reference* is the session's majority label. *Chip accuracy among called chips* counts every chip that received a *pass* or *fail* — whether the rule stopped at confidence  $\geq 0.9$  or exhausted the budget outside the inconclusive band; *false-confident rate* is the fraction of called chips whose call was wrong with posterior confidence  $\geq 0.9$ ; *inconclusive rate* is the fraction that exhausted the budget near 0.5. *Bad-region recall* (sampling simulations) is the fraction of the session's bad fields that the sampled set contains.

**Evaluation discipline.** Every model number in this report comes from a **session-disjoint** test set of 25 sessions (684 fields). These are the withheld half of each held-out session's fields (the controlled-leakage design sets the other half aside; it is never used for training), so a "test chip" is a random half of a session; §6.4 repeats the tool on all 1,377 fields of the same sessions. Where we also report the shipped split's number, it is labelled as such. Training hyperparameters were fixed

before the final evaluation. One decision-rule setting was not: the `min_fields` guard was read off these chips, and §6.2(c) re-selects it without them.

## 4. Audit

### 4.1 The published split leaks sessions

**Observation.** Parsing the archive's central directory shows the split is image-level: sessions are not held out. **Train contains 59 sessions, validation 51, test 57, and train  $n$  test = 57 of 59 sessions** — images of the same chip, acquired in the same session, appear on both sides.

**Controlled measurement.** A plain comparison of splits confounds leakage with differences in test composition, so we hold everything else fixed:

- the **test images** are identical in both arms (20 sessions, 494–712 fields depending on the seed);
- the **training-set size is identical** in the two arms (1,440–1,876 fields depending on the seed; the 200 validation fields are identical too and outside both training sets);
- the **model and budget are identical** (MobileNetV3-small [4], ImageNet-initialised, 6 epochs, 224 px);
- the only difference is whether the training set may use *the other images of the test sessions* (leaky) or must come from disjoint sessions (disjoint).

| arm                                           | accuracy                     | AUC                        |
|-----------------------------------------------|------------------------------|----------------------------|
| session-disjoint (mean $\pm$ s.d. over seeds) | <b>67.4%</b> ( $\pm$ 6.8 pp) | <b>0.753</b>               |
| same sessions in training (leaky)             | <b>75.2%</b> ( $\pm$ 3.3 pp) | <b>0.842</b>               |
| <b>inflation</b> (paired, 8 seeds)            | <b>+7.9 pp</b> [4.2, 11.5]   | <b>+8.9 pp</b> [6.7, 11.0] |

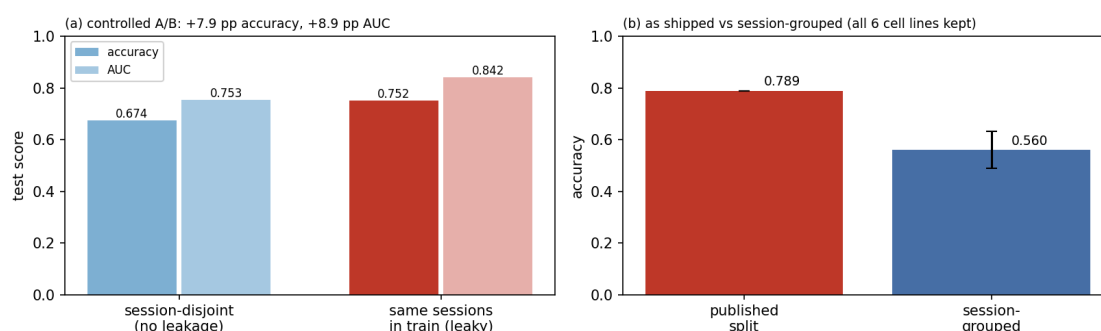


Figure 2. (a) controlled A/B (8 seeds; paired inflation +7.9 pp accuracy, +8.9 pp AUC; both gains are positive in every seed: accuracy 4.2–17.8 pp, AUC 6.7–15.0 pp); (b) the shipped split versus a session-grouped split.

For reference, the shipped split yields **78.9%  $\pm$  0.07 pp / AUC 0.873** (mean  $\pm$  s.d. over three seeds; the variation is tiny because all test sessions are seen in training), while a session-grouped split that keeps all six cell lines yields **56.0%  $\pm$  7.2 pp / AUC 0.684** — a 22.9 pp gap that mixes leakage with the small number of held-out sessions (six). We therefore treat the **controlled A/B as the headline number** and report the shipped/grouped pair only as context.

**Why this matters for the benchmark.** Any model compared on the shipped split is compared on a test set whose sessions it has already seen. Model *rankings* may survive (the leak affects all models similarly), but the *absolute* numbers, and any claim of "generalisation to new chips", do not.

## 4.2 Labels are clustered along the acquisition order

**Runs test.** For each session we take the label sequence in field order and count runs (maximal contiguous blocks of equal label). Under independence the expected run length is  $1 / (2p(1-p))$ ; with  $p \approx 0.56$  that is **2.03** fields. Observed: **6.08**. A Wald-Wolfowitz test is significant ( $p < 0.05$ ) in **23 of the 45 mixed sessions**, and 36/45 have negative  $z$  (clustered) — against 22.5 expected by chance. The most extreme session has  $z = -11.9$  (21 runs where 107.3 were expected, 215 fields).

| session | fields | good share | runs | expected runs | $z$    | $p$   |
|---------|--------|------------|------|---------------|--------|-------|
| 230425  | 215    | 0.55       | 21   | 107.3         | -11.93 | <1e-6 |
| 230419  | 124    | 0.07       | 2    | 16.0          | -10.71 | <1e-6 |
| 230320  | 229    | 0.77       | 27   | 82.5          | -10.34 | <1e-6 |
| 230529  | 216    | 0.43       | 48   | 106.6         | -8.18  | <1e-6 |
| 230317  | 100    | 0.73       | 10   | 40.4          | -7.78  | <1e-6 |
| 230315  | 94     | 0.55       | 13   | 47.5          | -7.23  | <1e-6 |
| 230517  | 90     | 0.60       | 12   | 44.2          | -7.11  | <1e-6 |
| 230314  | 50     | 0.44       | 3    | 25.6          | -6.57  | <1e-6 |
| 230424  | 108    | 0.51       | 21   | 55.0          | -6.57  | <1e-6 |

Table 1. The nine most clustered sessions, by  $z$  (the last two tie; 15 of the 45 mixed sessions have  $p < 1e-6$ ). Expected runs are what independence would give for the same good/bad counts.

**This is not a cell-type artifact.** A session can contain several cell types imaged in contiguous blocks, which would create clustering for trivial reasons. Restricting the test to *maximal contiguous same-cell-type stretches* (at least 8 fields and both labels present; 72 stretches from 41 sessions), **38 of 72 remain significant** (57/72 clustered).

**It is not duplicate frames either.** In the six largest sessions, downscaled correlation between consecutive fields is 0.58–0.74 (random pairs in the same sessions: 0.06–0.36; over the six largest and six smallest sessions, mean 0.54 vs 0.28), i.e. consecutive fields are spatially adjacent — but counting a new view whenever a field's correlation with the previous one falls below 0.95 gives **1,161 distinct views among 1,183 images (98%)**. The clustering is therefore not redundancy; **failures occupy contiguous regions of the chip**.

**Effective sample size.** With a session-level intraclass correlation of **ICC = 0.321** and an ANOVA-adjusted mean cluster size of  $m \approx 51$  fields, the design effect is  $1 + (m-1) \cdot \text{ICC} = 17.0$ , so for estimating a metric the benchmark carries the information of roughly **3,072 / 17  $\approx$  181 independent labels** (a model may still exploit correlated fields in ways an estimator cannot, so this is a bound on *estimation*, not on learning). The mean lag-1 autocorrelation is 0.32 (53% of sessions above 0.3).

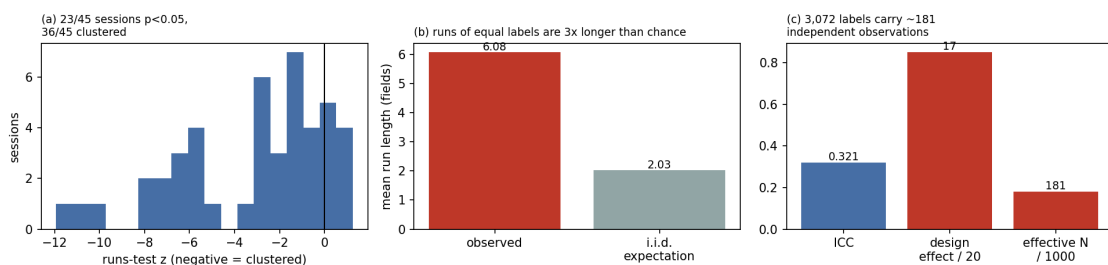


Figure 3. (a) runs-test  $z$  per session; (b) observed versus i.i.d. run length; (c) ICC, design effect and effective sample size.

Agreement degrades with culture age and is cell-line dependent:

| culture age | fields | single-field agreement |
|-------------|--------|------------------------|
| 0-1_days    | 852    | 0.805                  |
| 2-3_days    | 798    | 0.779                  |
| 4+_days     | 1199   | 0.676                  |
| 4_days      | 223    | 0.704                  |

| cell line | fields | single-field agreement |
|-----------|--------|------------------------|
| A549      | 775    | 0.710                  |
| CACO      | 346    | 0.792                  |
| HPMEC     | 1462   | 0.702                  |
| HSAEC     | 244    | 0.840                  |
| HUVEC     | 107    | 0.991                  |
| NHBE      | 138    | 0.826                  |

Table 2. Single-field agreement with the session's majority label, by culture age and cell line, counting every field once (74.1% over all 3,072 fields; the 78.8% in the text gives each session equal weight, see below). Late cultures are the hardest, and no cell line is uniformly easy.

**Consequence for labels, not just for statistics.** If the label were a property of the chip, a single field would predict the session's majority label almost perfectly. It does not: a randomly drawn field agrees with the session majority **78.8%** of the time, three fields 84.1%, five 87.4%, and even ten fields only **91.0%** (up to 50 random draws per session, so each session weighs about equally; counting every field once instead gives 74.1% for one field). Agreement is worse for late cultures (4+ days: 67.6% for a single field) than for day 0–1 (80.5%). The "quality" of a chip is therefore **a property of a region and of the sampling**, not of an image.

#### 4.3 Sampling policy: a comparison that does not survive the model in the loop

If failures are contiguous, how should fields be chosen? We first simulate three policies on the measured **label sequences** at a fixed budget  $k$  (300 trials per session):

- **random** —  $k$  distinct uniform fields;
- **window** — one contiguous run of  $k$  fields (a scan);
- **adaptive** — start at a random field; if it was bad, expand to an unsampled neighbour; if good, jump elsewhere.

| budget $k$ | random: acc / recall / err | window: acc / recall / err | adaptive: acc / recall / err |
|------------|----------------------------|----------------------------|------------------------------|
| 2          | 0.777 / 0.098 / 0.182      | 0.787 / 0.095 / 0.220      | 0.772 / 0.103 / 0.203        |
| 4          | 0.835 / 0.197 / 0.119      | 0.819 / 0.186 / 0.173      | 0.803 / 0.214 / 0.154        |
| 8          | 0.887 / 0.351 / 0.075      | 0.879 / 0.331 / 0.130      | 0.828 / 0.391 / 0.121        |
| 12         | 0.916 / 0.430 / 0.055      | 0.890 / 0.413 / 0.107      | 0.830 / 0.475 / 0.109        |
| 20         | 0.927 / 0.400 / 0.046      | 0.885 / 0.380 / 0.107      | 0.781 / 0.482 / 0.138        |

Table 3. Label-based policy simulation.  $err = |estimated - true\ bad\ fraction|$ ; a tied vote (even  $k$ ) is broken at random. Random looks best for the call from  $k = 4$  and adaptive best for localisation — and we do not believe this table.

This is exactly the practice criticised in §6.7, so we repeat the comparison with the deployed model's per-field probabilities on the 25 session-disjoint test chips (`audit/03c_policy_model_in_loop.py`):

| budget $k$ | random: acc / recall | window: acc / recall | adaptive: acc / recall |
|------------|----------------------|----------------------|------------------------|
| 4          | 0.791 / 0.335        | 0.785 / 0.321        | 0.772 / 0.323          |
| 8          | 0.789 / 0.300        | 0.818 / 0.311        | 0.776 / 0.296          |
| 12         | 0.800 / 0.393        | 0.818 / 0.389        | 0.840 / 0.397          |

Table 4. The same policies with the model in the loop. The ordering changes — adaptive, the label simulation's worst caller at  $k \geq 8$ , is nominally best at  $k=12$  — but nothing survives a multiple-comparison correction. A paired bootstrap over chips (10,000 resamples, `audit/03d_policy_bootstrap.py`) gives 95% intervals at  $k=12$  (13 chips) of  $[-0.08, -0.00]$  for random — adaptive,  $[-0.13, +0.07]$  for window — adaptive and  $[-0.13, +0.08]$  for random — window. The first only just excludes zero and rests on the 4 of 13 chips where the two policies differ at all. With a Bonferroni correction for the six differences tested (three pairs  $\times k = 8, 12$ ) no interval excludes zero — that one's upper bound is exactly 0 (Figure 4c).

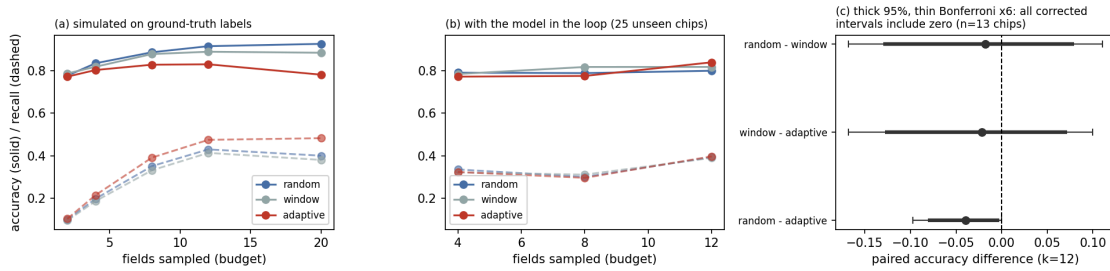


Figure 4. Policy comparison. (a) simulated on ground-truth labels; (b) with the model in the loop; (c) paired bootstrap differences at  $k=12$  — thick bars 95%, thin bars Bonferroni-corrected for six comparisons; every corrected interval includes zero, so no ranking is claimed.

**We therefore report no policy ranking.** With only 13–15 chips large enough for these budgets the comparison is underpowered, and the label-based ranking is an artefact of scoring policies against the very labels they were designed to sample.

What *is* established, and what the tool uses, is a single reproducible failure of the obvious default. Reading the **first**  $k$  fields in acquisition order with the sequential rule of §6.1 called a 100%-bad chip (session 230405) *pass* with 0.94 confidence: the per-field probabilities in the first third of that session are low (mean 0.255 over the first five fields, versus 0.935 over the last twenty). Sampling fields spread across the chip fixes that case (the same chip returns *fail*,  $P = 0.91$ ); on the 11 chips with at least 20 fields the accuracy difference (0.727 vs 0.636) is  $\pm 1$  chip and is not significant either. The justification for spread sampling is the observed failure, not a measured ranking.

#### 4.4 The same defect in a second organoid benchmark

To test whether the leakage we measure is a property of one dataset or of the field, we audited a second public organoid imaging benchmark: the **OCT organoid segmentation-and-tracking dataset** (zenodo.15783866, CC-BY-4.0; Branciforti et al., *Diagnostics* 2024 [5,6]). Its file names encode the acquisition group — `w<well>_d<day>_s<slice>.png` in train/val and `d<day>_p<plate>_w<well>_s<slice>.png` in test (all test images are plate 1; train/val names carry no

plate). If train/val come from the same plate — the names give no other one — the same (well, day) means the same organoids imaged in the same session. Reading the archive's central directory over HTTP range requests (no 4.9 GB download; `audit/06_oct_leakage.py`):

- train: 5,584 images in **8** (well, day) groups; val: 1,396 images in 2 groups; test: 6,980 images in 10 groups (each image is stored three times — original, manual mask, automatic mask; we count originals)
- **train n test = 4 groups** — well 2 at days 5, 7, 11 and 13; val n test = 1 group
- **40.0% of test images (2,792 of 6,980) belong to a (well, day) group that also appears in training**

For a *tracking* benchmark this is more severe than for classification: under that reading the same organoid instances, imaged in the same session at the same timepoint, appear on both sides of the split. The file names cannot rule out a second plate with the same well numbering; the paper's own split description would settle it. We report the structural overlap rather than a re-trained inflation number, because re-training their segmentation/tracking pipeline is out of scope here.

Two datasets, two independent instances of the same defect — group-level splits that leak — which is the empirical basis for the first rule of the corrected protocol (§5).

## 4.5 A hypothesis we tested and rejected: re-image vs discard

The dataset's own definition of `bad` mixes **technical artifacts** (bubbles, defocus, deformed walls) with **biological problems** (morphology/density off expectation) [1]. If these could be separated, the actionable output would be "re-image" (cheap, the chip survives) versus "discard" (expensive, re-imaging is useless). Two observations encouraged the attempt: isolated bad runs are blurrier and darker than good fields (single-field runs: mean absolute gradient 0.089 vs 0.126; dark fraction 0.137 vs 0.094 — bubbles and defocus), whereas long bad runs are sharper (0.150) and higher in contrast (s.d. 0.243 vs 0.199; `audit/04b_run_image_stats.py`); and visually, 4/4 sampled isolated-bad fields showed bubbles or defocus.

We constructed a target — a run of bad fields "recovers" if at least three of the next five fields are good; runs followed by fewer than five fields, including the 30 of 250 that reach the end of the session, count as not recovering (base rate 62.4%) — and tried to predict it from features available at decision time (run length, whether the run reaches the end of the session, focus, darkness, artifact scores), with **session-grouped cross-validation**.

| feature set                                                       | CV AUC                                   |
|-------------------------------------------------------------------|------------------------------------------|
| run length only                                                   | 0.476                                    |
| artifact signals only                                             | 0.381                                    |
| run length + artifacts                                            | 0.539                                    |
| all features                                                      | 0.613                                    |
| permutation null                                                  | mean 0.479, p95 <b>0.564</b>             |
| simple rule ( $\text{length} \leq 2 \Rightarrow \text{recover}$ ) | accuracy 0.560 vs base rate <b>0.624</b> |

**The decision rule is not supported.** The target is confounded by position (because failures are contiguous, the fields after a run are usually good, so "recovery" largely means "the run did not reach the end"), and — more fundamentally — **no re-imaging was ever performed in this dataset, so the causal question cannot be observed**. We therefore ship `pass/fail/inconclusive` and document the two-mode observation as an *open problem*, not a feature.

## 4.6 Aggregation rules are a sensitivity/specificity dial

If several fields are read, their scores must be combined. We compare three rules on per-field probabilities from a session-disjoint model, at a fixed field threshold chosen on validation, with the chip reference defined as before:

| budget k | mean: acc / sens / spec | max: acc / sens / spec | top-2: acc / sens / spec |
|----------|-------------------------|------------------------|--------------------------|
| 1        | 0.704 / 0.456 / 0.832   | 0.704 / 0.456 / 0.832  | 0.704 / 0.456 / 0.832    |
| 3        | 0.691 / 0.475 / 0.803   | 0.699 / 0.691 / 0.704  | 0.709 / 0.610 / 0.760    |
| 5        | 0.678 / 0.493 / 0.767   | 0.626 / 0.686 / 0.598  | 0.627 / 0.574 / 0.653    |
| 8        | 0.742 / 0.645 / 0.787   | 0.627 / 0.878 / 0.507  | 0.660 / 0.787 / 0.598    |
| 12       | 0.742 / 0.851 / 0.710   | 0.570 / 1.000 / 0.354  | 0.631 / 0.997 / 0.452    |

Table 5. Aggregation rules (224 px model, 2 seeds  $\times$  25 held-out sessions; 60 random k-field draws per chip, shared by all rules; chips with  $< k$  fields excluded).

mean maximises chip accuracy and keeps specificity high; max maximises sensitivity (it calls almost everything bad: specificity collapses to 0.35 at  $k=12$ ) and top-2 sits between. **The aggregation rule moves the operating point more than the per-field model does** — at  $k=12$ , accuracy ranges from 0.570 to 0.742 and sensitivity from 0.851 to 1.000 depending only on the rule. A QC deployment must therefore state its aggregation rule; "accuracy" without it is not a specification.

We also tested a rule that exploits the measured autocorrelation — smoothing the per-field scores along the acquisition order before aggregating. It did **not** help: on the shipped predictions a 3- or 5-field moving average changes chip accuracy by at most 0.007 at any budget from 4 to 12 fields ( $k=6$ : 0.792  $\rightarrow$  0.795 with a 3-field and 0.793 with a 5-field window; `audit/smoothing_test.py`). Reported as a negative result.

## 4.7 Metadata does not explain the labels

A logistic model on day, log seeding density, flow rate and cell line reaches only 0.562 in-sample accuracy, and apparent cell-line differences (Caco-2 68.5% bad vs NHBE 23.9%) dissolve on inspection: every cell line has sessions with a 0% bad rate and sessions with 100%. The label is dominated by **session-level** variation, consistent with §4.2.

## 5. Corrected evaluation protocol

From the audit, three rules follow for anyone training or benchmarking on this dataset:

1. **Split by session, not by image.** No session may contribute to two splits. Report the number of held-out sessions alongside the metric — with 59 sessions, a 20% held-out split is  $\sim 12$  sessions and the confidence interval on chip-level accuracy is correspondingly wide.
2. **Evaluate at the chip level.** Report (i) per-field accuracy/AUC and (ii) chip-level accuracy with the number of fields used, because per-field numbers do not answer the lab's question and are inflated by within-session correlation (effective  $N \approx 181$ , not 3,072).
3. **State the sampling policy.** Which fields were used, and how many. Policies differ in principle (§4.3) and we could not rank them with this benchmark's 59 sessions; a number is only interpretable together with the policy that produced it.

As a worked example, the scripts listed in §10 reproduce every number in this report from the raw dataset.

## 6. The tool

### 6.1 Design

`inference.py` takes the fields of one chip (filenames in acquisition order) and returns:

- **per-field  $P(\text{bad})$**  from a MobileNetV3-small (ImageNet-initialised, 384 px, 25 epochs, cosine LR, label smoothing, random-resized-crop/flip augmentation), trained on a **session-disjoint** split (1,495 train / 200 val / 684 test fields from 25 unseen sessions);
- a **chip call** — pass, fail, or inconclusive — from a Beta(1,1) posterior on the bad fraction, stopping when  $P(\text{chip bad}) > 0.9$  or  $< 0.1$ ;
- **how many fields were needed**, with fields chosen **spread across the chip** rather than the first  $k$ ;
- a **QC map** (field index vs  $P(\text{bad})$ ) and a `chip_report.json`.

**Model and training.** MobileNetV3-small [4], ImageNet-initialised, classifier head replaced with a 2-way linear layer. Input  $384 \times 384$  (the source images are  $2,056 \times 1,542$ ). AdamW (lr  $3e-4$ , weight decay 0.02), cosine schedule to lr/30, label smoothing 0.05, batch 32, 25 epochs ( $\approx 14$  s/epoch on one RTX 3090,  $\approx 6$  min total), RandomResizedCrop(0.7–1.0) + horizontal/vertical flips. Label smoothing and the modest capacity are deliberate: the labels are four-rater majorities, and 3,072 fields with an effective  $N$  of  $\sim 181$  do not support a large model. The checkpoint is shipped in `model/`.

### 6.2 Two bugs found during deployment, and how their fixes were chosen

Deploying the tool on real chips exposed two failure modes that a label-only simulation had hidden.

**(a) Reading the first  $k$  fields is wrong.** On a 100%-bad chip, walking the first fields in acquisition order stopped after six fields (five good predictions, one bad) and returned *pass* with 0.94 confidence. Because failures are contiguous (§4.2), the beginning of a session can lie inside a good region. Sampling fields spread across the chip fixed this specific case (the same chip now returns *fail* at 0.91).

**(b) The Beta posterior is overconfident on few fields.** With three good fields and no bad one,  $P(\text{chip bad}) = 0.0625$  — "94% confident" — which is indefensible when the per-field model is imperfect. Requiring a minimum of eight fields before a stop is allowed cut the false-confident rate from 25% to 13%.

**(c) The value 8 was read off the test chips, and it does not survive re-selection.** Both fixes above were found *on* the 25 test chips, and the minimum of eight was chosen from a sweep on them (Figure 5). To see how much of the test gain is selection, we re-selected it without the test chips (`audit/09_inner_cv_minfields.py`): five-fold session-grouped cross-validation over the 34 non-test sessions with the shipped training recipe, each session cut into two half-chips like the test chips (68 chips), and a criterion fixed before looking — highest chip accuracy with every chip called, ties to fewer fields.

| on 68 non-test chips (out-of-fold) | min 1 | min 8 | min 12 | all fields |
|------------------------------------|-------|-------|--------|------------|
| chip accuracy, every chip called   | 0.779 | 0.765 | 0.765  | 0.765      |

| on 68 non-test chips (out-of-fold) | min 1 | min 8 | min 12 | all fields |
|------------------------------------|-------|-------|--------|------------|
| confident-but-wrong chips          | 11    | 10    | 8      | —          |
| fields per chip                    | 5.3   | 8.1   | 9.9    | 24.9       |

The criterion selects **min 1**, whose test result is **0.680** with every chip called (0.708 on the 24 it calls, 6.8 fields). The guard's accuracy gain on the test chips (0.680 → 0.800) is therefore not reproduced on independent chips and should be read as optimistic; on those chips the guard only trims confident errors (11 → 8 of 68 at min 12) for one chip of accuracy. What *does* replicate is the efficiency claim: at every minimum the rule matches reading every field (0.765–0.779 versus 0.765) with 5.3–9.9 fields instead of 24.9, and in the deferring mode it is 0.806 on the 62 of 68 chips it calls at min 8 (0.800 on 65 at min 1) — close to the 0.826 measured on the test chips. We keep min 8 in the tool for the a-priori reason in (b) — three agreeing fields cannot justify 94% — and report the test-chip numbers at 8 below with this caveat. The same re-selection picks the plain cap of Table 6 as **20**, not 12 (0.765 on the 68 chips, 0.760 on the test chips with 12.6 fields), and reading the first fields instead of spread ones is no better there (0.750 vs 0.765 at min 8): the spread order is kept for the failure in (a), not for a measured gain.

### 6.3 Measured performance (25 unseen chips, 684 fields)

| metric                                                                                         | value                                                          |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| per-field accuracy / AUC                                                                       | 0.734 / 0.791                                                  |
| chip accuracy, all fields, mean                                                                | 0.80                                                           |
| <b>chip accuracy among called chips</b> (spread fields, min 8 — tuned on these chips, §6.2(c)) | <b>0.826</b> (95% Wilson CI 0.63–0.93, n=23)                   |
| <b>false-confident calls</b> (conf $\geq 0.9$ and wrong, of 23 calls)                          | <b>13%</b> (3/23)                                              |
| inconclusive (budget exhausted near $P = 0.5$ )                                                | 8%                                                             |
| mean fields used                                                                               | <b>9.5</b> (a plain cap of 12 spread fields: 9.0 fields, 0.80) |

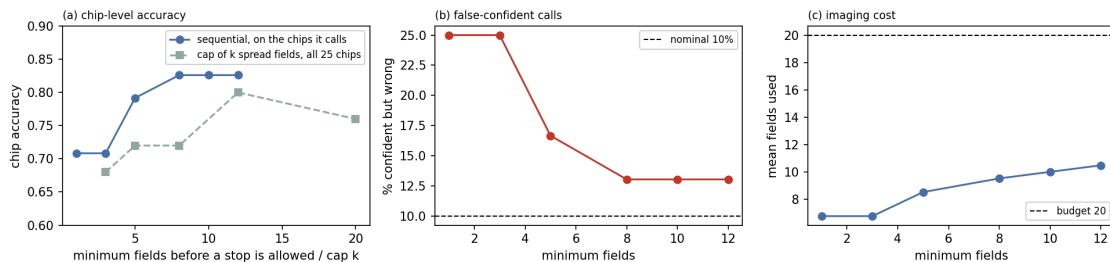


Figure 5. Tool behaviour versus the minimum-fields guard, swept on the 25 test chips (the sweep the value 8 was read from; §6.2(c) re-selects it without them). In (a) the sequential curve is accuracy on the chips it calls (23–24 of 25); the grey curve is a cap of  $k$  spread fields on all 25.

**Capacity is not the bottleneck.** We trained a backbone with 2.8× the parameters (MobileNetV3-large, same 384 px, same split and schedule; batch 8 instead of 32 to fit memory). Test field accuracy moved from 0.734 to 0.746 and balanced accuracy from 0.733 to 0.747, while AUC *fell* from 0.791 to 0.788 — i.e. a larger model buys nothing here (audit/perfield\_model.py --arch large). Together with the session-level shift in §6.5 (almost every unseen session sits outside the training feature distribution) this suggests the ceiling is set by the labels and by chip-to-chip appearance, not by model capacity. We therefore ship the smaller model, which is cheaper to run; with the same stopping rule the larger model uses the same 9.5 fields and is not more accurate on the chips it calls (0.792 vs 0.826).

## 6.4 Field efficiency: the same accuracy with 2.9× fewer fields

The naive way to use a per-field model is to read every field of the chip and average. On the 25 test chips that costs **27.4 fields per chip for 0.800 accuracy**. The shipped rule — fields spread across the chip, Beta posterior, stop at confidence 0.9, minimum 8 fields — reaches **0.800 on all 25 chips using 9.5 fields (2.9× fewer)**, and when it is allowed to defer the chips it cannot resolve it reaches **0.826 on the 23 it calls, deferring 8%** (`audit/07_efficiency.py`):

| policy                                                  | fields per chip | chips called | accuracy     |
|---------------------------------------------------------|-----------------|--------------|--------------|
| all fields (mean of every field)                        | 27.4            | 25/25        | 0.800        |
| fixed k = 12, spread (chips with $\geq 12$ fields only) | 12.0            | 13/25        | 0.846        |
| cap of 12 spread fields (all fields if fewer)           | 9.0             | 25/25        | 0.800        |
| <b>sequential, force</b> (min 8)                        | <b>9.5</b>      | <b>25/25</b> | <b>0.800</b> |
| <b>sequential, abstain</b> (min 8)                      | <b>9.5</b>      | 23/25        | <b>0.826</b> |

Table 6. Field efficiency. The fixed-k row is evaluated only on the 13 chips that have at least 12 fields, so it is not comparable with the others; the cap and sequential rows use every chip. A plain cap of 12 fields is as frugal as the sequential rule, but the right cap is only known after the fact (a cap of 8 gives 0.720, a cap of 20 gives 0.760; selected without the test chips the cap would be 20, §6.2(c)). The min-8 rows share the same caveat: without the test chips the minimum would be 1, which scores 0.680 here; the rule's contribution is the per-chip confidence, the pass / fail / inconclusive call and the option to defer, not a further cut in fields.

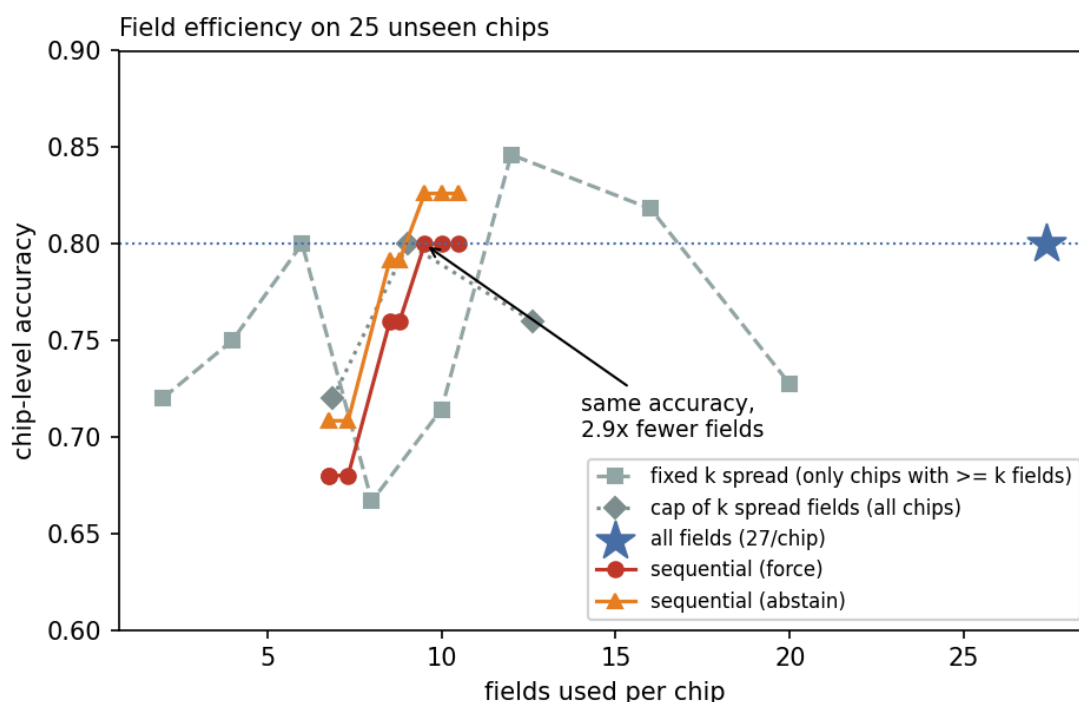


Figure 6. Accuracy against the number of fields used per chip.

For a lab imaging plates of chips this is the practical number: the decision costs roughly a third of the microscope time of the read-everything workflow, and a further 8% of chips are flagged "needs a human" instead of being guessed.

**Full sessions.** On all fields of the same 25 held-out sessions (1,377 fields, 55.1 per chip, reference = full-session majority; `audit/08_full_sessions.py`) the saving grows and the accuracy drops: reading every field gives 0.76, the shipped rule gives **0.76 with 9.0 fields (6.1× fewer)**, but it

defers no chip and 6 of its 25 calls are confident and wrong (min 10 or 12: 0.80, 11–12.6 fields). The half-session numbers above are therefore the favourable end.

The tool ships this as a **plate mode** (`inference.py --plate <folder>`; one subfolder per chip), which returns a ranked triage table. On the 25 test chips it spends **238 of 684 fields (65% saved)** and returns 6 *fail*, 2 *inconclusive* and 17 *pass*, worst first.

## 6.5 Where the tool fails

The model's own confidence does not flag chip-level failures: all four wrong calls in Table 7 were made at confidence 0.89 or higher, three of them above the 0.9 stopping threshold. The clearest case is session 230405 (69 fields, **100%** labelled bad): the model is right on 82.6% of its fields individually, yet the per-field probabilities in the first third of the session are low (mean 0.255 over the first five fields versus 0.935 over the last twenty), so an unguarded rule stopped early with a confident *pass*.

| chip (session) | fields | true bad share | per-field accuracy |
|----------------|--------|----------------|--------------------|
| 220706         | 5      | 100%           | 0.200              |
| 220718         | 4      | 100%           | 0.750              |
| 220721         | 3      | 100%           | 1.000              |
| 230405         | 69     | 100%           | 0.826              |
| 230403         | 6      | 83%            | 0.167              |
| 230314         | 25     | 64%            | 0.720              |
| 230512         | 44     | 61%            | 0.795              |
| 230529         | 108    | 59%            | 0.769              |
| 230523         | 55     | 55%            | 0.582              |
| 220606         | 10     | 50%            | 0.200              |
| 230425         | 107    | 49%            | 0.673              |
| 230315         | 47     | 47%            | 0.638              |
| 230215         | 12     | 42%            | 0.583              |
| 220506         | 7      | 29%            | 0.857              |
| 220715         | 7      | 29%            | 0.429              |
| 230123         | 13     | 23%            | 0.692              |
| 220515         | 6      | 17%            | 0.833              |
| 230109         | 53     | 13%            | 0.792              |
| 230214         | 24     | 12%            | 0.667              |
| 220502         | 4      | 0%             | 1.000              |
| 220620         | 6      | 0%             | 1.000              |
| 220716         | 4      | 0%             | 0.750              |
| 230119         | 35     | 0%             | 0.971              |
| 230316         | 22     | 0%             | 1.000              |
| 230321         | 8      | 0%             | 1.000              |

Table 7. The 25 unseen test chips, sorted by the true share of bad fields. The deployed rule (spread fields, min 8, conf 0.9) makes four wrong calls — 220706 (100% bad, 5 fields) and 230403 (83% bad, 6 fields) called pass with confidence 0.89 and 0.99, 230314 (64% bad) pass at 0.91, 230425 (49% bad) fail at 0.93 — and returns inconclusive on 220606 and 230529. Chips with fewer than 8 fields never reach the min-8 guard; they account for 2 of the 4 wrong calls (2 of 10 such chips, versus 2 of 15 larger ones).

**Out-of-distribution detection failed.** We tested two standard detectors against the training fields: a Mahalanobis distance on four image statistics (mean, standard deviation, mean absolute gradient as a focus proxy, dark fraction) and a Mahalanobis distance on the model's 576-dimensional pooled backbone feature, each alarming above the training 99th percentile (`audit/10_ood_check.py`). The image-statistics detector flags none of the 25 unseen chips. The feature-space detector flags 21 of 25 when a chip is scored by its median field distance (almost every unseen session looks shifted) and 4 of 25 when scored by its mean feature. **None of the three flags session 230405**, the worst failure. We therefore do not present an OOD alarm as a safety feature; the distance is exposed as a diagnostic only.

**What this means in practice.** The tool is usable as a *screening* aid with a human in the loop, not as an autonomous gate. Its measured error rate (13% confident-but-wrong) is the number to quote in any deployment decision.

## 6.6 Generalisation to a cell line the model has never seen

The session-disjoint protocol of §6.3 keeps all six cell lines in training. To measure a stricter setting we hold out one cell line entirely — test = all of its fields, train = the other five with any session containing the held-out line removed — and retrain the deployed configuration (`audit/05_lolo_cellline.py`):

| held-out cell line                     | test fields (sessions) | accuracy     | balanced acc | AUC          |
|----------------------------------------|------------------------|--------------|--------------|--------------|
| A549                                   | 775 (24)               | 0.712        | 0.661        | 0.678        |
| CACO                                   | 346 (17)               | 0.512        | 0.611        | 0.666        |
| HPMEC                                  | 1462 (29)              | 0.589        | 0.605        | 0.659        |
| HUVEC                                  | 107 (4)                | 0.879        | 0.874        | 0.930        |
| NHBE                                   | 138 (6)                | 0.601        | 0.499        | 0.630        |
| HSAEC                                  | 244 (21)               | 0.730        | 0.723        | 0.753        |
| <b>mean of the six</b>                 | 3,072 (101)            | <b>0.670</b> | 0.662        | <b>0.719</b> |
| in-distribution (all lines seen, §6.3) | 684 (25)               | 0.734        | 0.733        | 0.791        |

Table 8. Leave-one-cell-line-out. The spread between cell lines (AUC 0.630–0.930) is larger than the effect of any modelling choice we tested — a backbone with 2.8× the parameters or 512 px inputs change nothing (§6.3).

**Deploying on a new cell line costs about seven AUC points on average and up to 22 accuracy points** (Caco-2), and the per-line variation dominates. This is the strictest evaluation we have; we keep the session-disjoint protocol as the headline because it matches how such a tool would be used (a new chip of a known cell line), but any deployment to an unseen line needs re-calibration on that line.

## 6.7 Label-only simulation overstates deployed performance

A common practice — which we followed first — is to simulate a decision policy on **ground-truth labels**. Our first one (random order, no minimum, all 59 chips) gave 92.8% chip accuracy with 6.6 fields — not comparable with the deployed 82.6%, since rule and chips differ. Like for like, we run the *deployed* rule on the *same* 25 held-out chips with ground-truth field labels and with the model's calls (`audit/label_vs_model_same_rule.py`, Figure 7). Without a minimum, labels give  $21/24 = 0.875$  with one confident-but-wrong chip and the model  $17/24 = 0.708$  with six; with the shipped 8-field minimum, labels give 0.875 (8.5 fields) and the model  $19/23 = 0.826$  (9.5 fields, three confident-but-wrong). Per-field errors (AUC 0.791) let an unguarded rule stop early on a wrong run of calls; the 8-field minimum closes most of that gap on these chips — the chips it was tuned on (§6.2(c)).

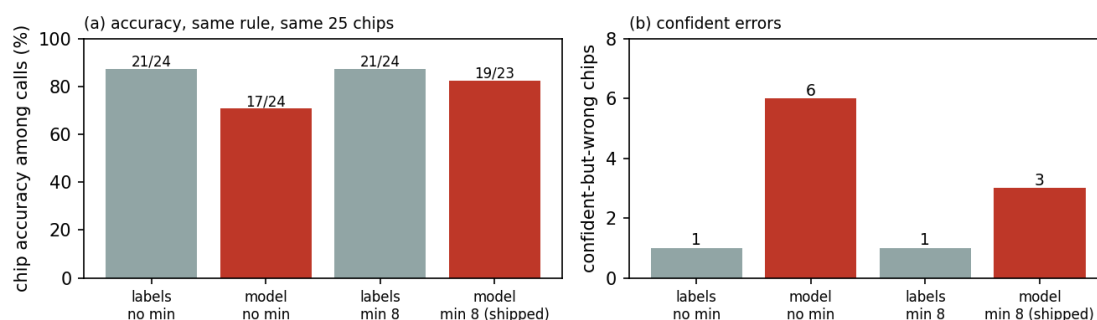


Figure 7. The same rule on the same 25 chips: ground-truth field labels versus model calls.

**Recommendation for the field:** evaluate QC decision policies **with the model in the loop**. A label-only simulation measures the policy, not the system.

## 6.8 Interactive demo

A browser-side build of the tool (ONNX Runtime Web, no server) is available at <https://taewoong23-ooc-chip-qc-demo.static.hf.space/index.html>. The bundled example chips use the reference probabilities computed by the Python implementation, so the demo reproduces the numbers in this report exactly; user uploads are scored live in the browser by an ONNX export of the same checkpoint (largest difference in  $P(\text{bad})$  over the 44 example fields:  $8.5e-06$ ; `audit/12_onnx_parity.py`). The page is static, so it stays available throughout the judging period without any server running.

## 7. Application value

**Laboratory automation.** The daily QC pass is the most repetitive imaging task in a chip lab: every chip is looked at, most are fine, and the decision is made by eye. The tool returns a ranked plate triage — on 25 test chips it flags 6 failures and defers 2, while spending 238 of 684 fields (65% saved) — so a human looks only where it matters, and the imaging budget follows the risk.

**Data assetisation and standardisation.** The audit's most transferable result is that a chip imaging dataset's *information content* is not its file count: 3,072 fields carry roughly **181 independent labels**, and a published split that leaks acquisition groups can inflate accuracy by **7.9 pp**. Any organisation building chip data assets — or training models on them — needs exactly this kind of measurement to know what its data is worth and when a reported number can be believed. The corrected protocol (session-level splits, chip-level metrics, explicit sampling policy) is dataset-agnostic and applies to any group-structured imaging corpus.

**Drug evaluation and toxicology.** QC is the gate in front of every downstream readout. A chip-level QC layer — with an explicit *inconclusive* outcome instead of a guess — can screen failing chips out before they reach dose-response curves (at the measured error rate: 3 of 23 calls confident and wrong, so a human confirms), and makes the provenance of each excluded chip auditable.

**Toward chip digital twins.** A digital twin needs a state estimate of the physical system at each time point. The tool's output is a first version of that at the culture level: a pass/fail state with a posterior confidence plus a spatial map of where the culture is degrading, and a measured cost (9.5 fields) for obtaining it. Feeding such state estimates into a model of the culture over time is the natural next step.

**What we do not claim.** No wet-lab validation, no biological or clinical validity, one dataset for the tool, and 3 of 23 calls on unseen chips are confident ( $\geq 0.9$ ) and wrong (§9). The contribution is a *measurement and decision layer* whose error rate is stated, not a biological finding.

## 8. Discussion

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**For benchmark authors.** The three defects we measure are cheap to check and were not checked here: a session-level (or patient-level, or chip-level) split; a design-effect estimate for the label; and a per-image agreement statistic between a single sample and the group's consensus. We suggest reporting all three alongside the usual accuracy, because they change what the number means. In this benchmark the effective sample size is 6% of the nominal one, and a model trained on the shipped split is evaluated on chips it has already seen.

**For laboratories.** The practical consequence of clustered failures is that *which* fields are imaged matters: the same rule reads the same chip correctly or not depending on whether it looks at the beginning of the acquisition order or across the whole chip. We could not establish a general ranking between random, scan and adaptive sampling with 59 sessions (§4.3), so the tool uses spread sampling to avoid the specific failure we observed rather than as an optimised policy. Its cost (9.5 fields on average) is modest; its error rate (13% confident-but-wrong) is the number that matters in a deployment decision.

**For method developers.** The largest single lesson is in Figure 7: without a minimum-field guard the same rule on the same chips is 0.875 accurate on labels and 0.708 with the model in the loop. Per-field errors interact with early stopping. QC papers that report policy numbers from label simulations should be read with this in mind, and we recommend model-in-the-loop evaluation as the default.

**Open problem.** The dataset's own definition of bad mixes technical artifacts with biological failure, which suggests the actionable output should be "re-image" versus "discard". We tried and failed to validate that mapping (§4.5): the causal question cannot be observed without actual re-imaging. A dataset that images each chip twice — before and after a re-imaging attempt — would settle it, and would be a small, valuable addition to this benchmark.

## 9. Limitations

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1. **13% confident-but-wrong** on unseen chips: 4 of 23 calls are wrong, 3 of them with  $\geq 0.9$  confidence. The stated confidence is a posterior under an independence assumption, not a calibrated probability: it averages 0.93 over the 23 calls, while 0.826 of them are correct. This is a research prototype, not a validated instrument; a chip should not be discarded on its output alone.
2. **No reliable out-of-distribution detector.** We tested image-statistics distance and feature-space Mahalanobis distance against the training chips; both failed to flag the worst case (the

100%-bad chip called *pass* with 0.94 confidence). The tool exposes the distance as a diagnostic only.

3. **The re-image/discard recommendation is not validated** (§4.5) and is not part of the tool's output.
4. **One dataset, 25 test chips.** Chip-level metrics carry wide confidence intervals; we report them as point estimates with the split size. Each test chip is half of a held-out session; on the full sessions the rule is 0.76 accurate with 6 of 25 confident-but-wrong (§6.4). No wet-lab validation was performed, and we make no biological or clinical claim — the claim is about *measurement validity*.
5. **Sampling policies could not be ranked.** The label-based comparison (§4.3, Table 3) is underpowered when repeated with the model in the loop (13–15 chips; no paired difference survives a correction for six comparisons). We report the policy we use and the failure that motivates it, not a ranking.
6. **Labels come from a four-rater majority**, so the ceiling of any model on this benchmark is the agreement among experts, which is not public. Our numbers are therefore a lower bound in that specific sense, while the leakage inflation is an upper-bound problem.
7. **Generalisation to a new cell line is untested in deployment.** Held-out cell lines cost ~7 AUC points on average (0.791 → 0.719) and the spread between lines dominates every modelling choice we tested (§6.6). The tool is validated for cell lines it has seen.
8. **The acquisition order is used as a proxy for spatial order.** Consecutive fields are highly correlated ( $r = 0.58\text{--}0.74$  vs  $0.06\text{--}0.36$  for random pairs in the six largest sessions), which supports the proxy, but the exact stage geometry is not public.
9. **The minimum-fields guard was tuned on the test chips.** The shipped value 8 comes from a sweep on the 25 test chips; re-selected on the 34 other sessions it would be 1 (test accuracy 0.680 instead of 0.800 with every chip called). The headline 0.826 / 13% are therefore optimistic by an unknown amount; the field saving is not (§6.2(c)).

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## 10. Reproducibility

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```
pip install -r requirements.txt          # torch, torchvision, numpy, Pillow, scipy,
scikit-learn,                           # matplotlib, openpyxl, gradio, markdown,
playwright

# data (not redistributed here) – the zip holds a top-level OOC_image_dataset/ folder
mkdir -p ../data
curl -L -o ooc.zip
"https://zenodo.org/api/records/10203721/files/OOC_image_dataset.zip/content"
python3 -c "import zipfile; zipfile.ZipFile('ooc.zip').extractall('../data')" #
unzip(1) fails on this zip64
rm ooc.zip                               # lands at ../data/OOC_image_dataset/ (or set
OOO_DATA)

python3 audit/leakage_experiment.py      # published split: 57/59 sessions shared;
shipped vs grouped (§4.1)
python3 audit/leakage_controlled.py      # controlled A/B, +7.9 pp / +8.9 pp over 8 seeds
 (§4.1)
python3 audit/01b_leakage_ci.py          # paired 95% CIs over the 8 seeds (§4.1)
python3 audit/structure_probe.py         # runs test, Table 1; metadata model (§4.2,
 §4.7)
python3 audit/02c_celltype_control.py    # runs test within same-cell-type stretches
 (§4.2)
python3 audit/02d_redundancy.py          # consecutive vs random field correlation,
```

```

distinct views (§4.2)
python3 audit/block_structure.py          # ICC 0.321, design effect 17, effective N  $\approx$  181 (§4.2)
python3 audit/label_sufficiency.py        # single-field agreement, Table 2 (§4.2)
python3 audit/adaptive_sampling.py        # policy comparison on labels, Table 3 (§4.3)
python3 audit/03c_policy_model_in_loop.py # ... with the model in the loop, Table 4 (§4.3)
python3 audit/03d_policy_bootstrap.py     # paired bootstrap CIs over chips, Figure 4c (§4.3)
python3 audit/06_oct_leakage.py           # second benchmark, 40% of test images leak (§4.4)
python3 audit/recovery_test.py            # re-image vs discard, negative result (§4.5)
python3 audit/04b_run_image_stats.py      # focus/darkness of isolated vs long bad runs (§4.5)
python3 audit/aggregation_experiment.py   # mean / max / top-2 aggregation, Table 5 (§4.6)
python3 audit/smoothing_test.py          # smoothing along acquisition order, negative (§4.6)
python3 audit/perfield_model.py           # trains the shipped per-field model (§6.1; --arch large, --size 512)
python3 evaluate.py                      # deployed-tool numbers, Figure 5 (§6.3)
python3 audit/07_efficiency.py            # field efficiency, Table 6 (§6.4)
python3 audit/08_full_sessions.py        # the tool on full sessions (§6.4)
python3 audit/09_inner_cv_minfields.py    # min-fields guard re-selected without the test chips (§6.2(c))
python3 audit/10_ood_check.py            # OOD detectors on the 25 unseen chips (§6.5)
python3 audit/11_per_chip_calls.py        # per-chip calls, Table 7; the first-k failure (§4.3, §6.2(a))
python3 audit/05_lolo_cellline.py         # leave-one-cell-line-out, Table 8 (§6.6)
python3 audit/12_onnx_parity.py          # browser (ONNX) model vs PyTorch checkpoint (§6.8)
python3 audit/label_vs_model_same_rule.py # labels vs model, same rule and chips (§6.7)
python3 audit/stopping_rule.py           # the original label-only stopping simulation (§3, §6.7)
python3 figures.py                       # every figure in this report
python3 demo/app.py                      # interactive demo

```

Every number in this report is written to `results/*.json` by the script that produced it; `python3 check_numbers.py` fails if a number in REPORT/README cannot be traced to one of them.

## 11. Related work

**Leakage in benchmarks.** Split leakage is a known and actively studied class of problem: Ramos et al. audit leakage across seven visual datasets and measure its effect on downstream evaluation [7]; whole-slide-image multimodal benchmarks have been shown to leak at patient *and* institution level, with a measurable accuracy gap between leaked and audit-clean cases [8]; OCT classification was shown to be inflated by improper splitting [9]. Tooling exists for near-duplicate detection and grouped splitting [10,11]. Our finding is a new instance of this class, with the magnitude measured in an OoC QC benchmark by a controlled A/B.

**Clustered labels and effective sample size.** Treating images as correlated clusters when computing power or sample size is standard in imaging study design [12], and effective sample size under autocorrelation is textbook. We apply these tools to field-level QC labels and quantify the consequence for a benchmark ( $\approx$ 181 independent labels).

**Microscopy QC.** Automated QC for high-throughput microscopy has its own benchmarks and methods (e.g. AutoQC-Bench, 2025 [13]); classical QC metrics for microscopy images are long established [14]. Our contribution is complementary: not a new QC metric, but a measurement audit and a sampling/decision policy for chip-level QC.

**Organoid/OoC imaging benchmarks.** Several organoid imaging datasets exist (detection, segmentation, tracking), including multi-rater annotation [15]. None, to our knowledge, ship a chip-level QC protocol; the field-dependence we measure is a property of this label definition.

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## 12. Conclusion

A public benchmark for organ-on-a-chip quality control reports per-image accuracy on a split that leaks sessions (+7.9 pp measured by controlled A/B, 8 seeds), with labels that are clustered along the acquisition order (ICC 0.321; effective  $N \approx 181$  of 3,072) and field-dependent (a single field agrees with the session majority 78.8% of the time). Correcting the protocol — session-grouped splits, chip-level evaluation, explicit sampling policy — and sampling fields spread across the chip yields a tool that reaches 82.6% chip-level accuracy on the chips it calls with 9.5 fields on average, and says "inconclusive" when it cannot decide (the 8-field minimum behind that number was tuned on the test chips; re-selected without them the rule scores 0.680, §6.2(c)). Two negative results bound the claims: the re-image/discard distinction is not supported by this data, and no reliable OOD detector was found. Simulating the same stopping rule on ground-truth labels overstates its performance with the model in the loop (0.875 vs 0.708 without a minimum-field guard), a caution that generalises beyond this benchmark.

**Future work.** (i) Extend the model-in-the-loop versus label-only comparison to a second dataset and modality (a public patient-derived-organoid drug-response dataset is a candidate); (ii) a calibrated stopping rule that accounts for the measured design effect; (iii) prospective validation of the two-mode (transient vs persistent) hypothesis with actual re-imaging.

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## References

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